21-242 Recitation 12

Matrices — Q2. Every matrix is some [T]

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Show that every matrix A can be written as $A = [T]_{\mathcal{CB}}$ for some linear transformation T and bases $\mathcal{C}$ and $\mathcal{B}$.

Course-level solution

Let $A \in \mathbb{F}^{m\times n}$. Take $V = \mathbb{F}^n$, $W = \mathbb{F}^m$, let $\mathcal{B}$ and $\mathcal{C}$ be the standard bases, and let $T = T_A$ (the map $\vec{x} \mapsto A\vec{x}$; defined in Q5). The jth column of A is $Ae_j = T_A(e_j)$, whose standard coordinates are the jth column of A. Thus $[T_A]_{\mathcal{CB}} = A$.

Later-course solution (using ideas from the rest of the handout)

You may also keep V and W arbitrary of dimensions n and m, choose any bases $\mathcal{B}, \mathcal{C}$, and take the unique T with $[T]_{\mathcal{CB}} = A$ (Q1). So every matrix arises in every pair of spaces of the right dimensions.